SolariNet — Operator Quick Reference

Dashboard-first field reference. · Historical snapshot 2026-07-07; current state note added 2026-10-21.

Historical operational details — verify before acting. This guide records a 7 July snapshot; its host versions, service locations, and recovery procedures were not revalidated for this edition. Steel is the inventory-labeled ZimaBlade on laboratory switch port 12; it is unreachable and is not currently advertised as a DNS server. Chlorine is the Khadas VIM4 at 10.7.0.10 and now runs an authoritative-only BIND secondary for 109 zones (forward, reverse, bare-name, and RPZ); canonical zone hashes match Xenon. It refuses recursion. See the current stabilization baseline for checked state and release gates.

▶ Start here — the dashboard

https://dashboard.akoria.net/

TLS self-signed · Local login · Keycloak OIDC (akoria.org)

Two ways in: a **local admin** account, or **Sign in with Keycloak** — your AD identity on `akoria.org`. Domain Admins map to dashboard **admin**; everyone else read-only.

WHAT EACH AREA IS FOR

Assets / Inventory

The live CMDB view. Every discovered + adopted host, its IPs, OS, role, and monitoring state. This is the human face of the **System of Record**.

Discovery

Newly-seen hosts on the network (nmap/mDNS sweep). You **Adopt** the ones worth tracking — that promotes them into the SoR and starts monitoring them.

Monitoring / Health

~49 targets, app-layer aware: not just ping, but real LDAP / MySQL / AMQP / DNS / HTTP checks on the infra that matters.

Topology

Combined network + monitoring map — nodes, interconnects, and their health, from real LLDP-over-SNMP.

Host health & alerts

Every client reports host-local telemetry each cycle — fs-readonly, missing block devices, SMART, failed systemd units, dmesg-critical lines — independent of whatever app is running there. fs-readonly / missing-device / SMART-fail are **crit**, sustained immediately (no debounce); failed-units and dmesg-crit are warn. A **dead-man's-switch** also watches the monitor itself: a node silent past 3× its sample interval pages on its own.

THE ONE THING TO UNDERSTAND

The dashboard writes the System of Record, and the SoR drives everything else — automatically. You never hand-edit a zone file, a Pi-hole list, or AD again. Make the change once, in the dashboard, and it fans out.

Dashboard edit → **SoR** (MariaDB, cesium) → CDC outbox → `sor.events` (RabbitMQ) → appliers fan out to:

BIND	`akoria.net` forward/reverse	xenon · auto rndc reload on real change
Active Directory	`akoria.org`	radium · safe deletes (never touches AD infra records)
Pi-hole	helium + mercury `custom.list`	reload on change

And the reverse: a host you **Adopt** in Discovery flows the same way — into the SoR, then out to DNS — with no extra step.

COMMON TASKS

Add a host to DNS

1. Find it in **Discovery** (or add in Assets).
2. Click **Adopt** / set its name.
3. Done — A + PTR appear in `akoria.net` within seconds; AD + Pi-hole follow.

Rename or remove a host

1. Edit the name (or **Remove**) in **Assets**.
2. SoR updates; DNS re-renders and reloads automatically.
3. Removal soft-deletes → record drops from all three DNS targets.

Start monitoring a service

Adopt with the ports you care about. Ports `389` / `3306` / `5672` auto-upgrade from a bare TCP check to a real **LDAP** / **MySQL** / **AMQP** app-layer probe. CLI equivalent:

`ADOPT disc=<id> services=389,3306 heartbeat=1` → `run/solariCtl.sock`

Get alerts

Alerts publish to `notify.events` and fan out by severity:

info	log + MQTT only (quiet)
warn	+ **Apple** (iMessage → all your devices)
crit	+ **Apple**

Apple channel relays via **hydrogen** → iCloud → iPhone/iPad/Watch/Mac.

ENDPOINTS & HOSTS

SERVICE	ADDRESS	NOTES
Dashboard	`dashboard.akoria.net`	Apache, TLS (SNI :443), local + OIDC
System of Record	`cesium 10.1.0.200:3306`	MariaDB, 30-table CMDB/IPAM
SoR replica	`benzene 10.5.2.50:3307`	real-time mirror (podman)
RabbitMQ	`benzene 10.5.2.50:5672` · UI `:15672`	vhost `solari`
MQTT (Mosquitto)	`benzene 10.5.2.50:1883` · ws `:9001`	
DNS — internal	`xenon` BIND	`akoria.net` authoritative
DNS — AD	`radium 10.1.0.10`	`akoria.org`, Samba AD DC + Keycloak
Pi-hole	`helium 10.0.0.254`, `mercury 10.0.0.253`	ad-block + local records
Apple-notify relay	`hydrogen 10.0.1.50`	Mac; Messages signed in as jason.kay@mac.com

THE STATUS PANEL

xenon `/api/panel` → **lithium** (`solariPanel`, polls 5s) → USB-CDC → **the wall** (Galactic Unicorn, 53×11)

A 583-LED matrix that cycles 12 screens across 4 themes. It renders what xenon computed — score, alarm, counts — and owns everything else itself: rotation, brightness, its own history, and the acknowledge state of the current alarm. Drive it from **Panel** on the dashboard, or from the nine buttons on the unit.

Reading it at a glance

slate, dim, still	operational
amber, pulsing	degraded
red	down (faster pulse = tier 0)
slate, dashed	unknown — enrolled, reporting nothing
azure	**never a status** — chrome, labels, zero counts

A nearly-dark panel is a healthy fleet, not a dead one. Every screen carries something that moves; that motion is the liveness check.

The four themes

A Abstract	state lattice · flow gates · MQ/SNMP lanes
B Text	condition counts · throughput · load & latency
C Charts	load histogram · live traces · pool load
D Instruments	reachability · pool bars · ambient field

Boots to **D1, Pool bars**. Each screen can be taken out of rotation, or weighted `¼×` – `10×` against the 3/6/30s base dwell — at a 30s base, 5× holds for 150s and ½× for 15s.

When it alarms

1. Both edge columns go red, the subject prints on row 1, the detail scrolls on row 6.
2. A three-note tone sounds, and **repeats every 60s** until acknowledged. Auto-silences after 5 minutes — tone only.
3. Four pixels flash top-right: the **beacon**. It survives sleep and dimming.
4. Acknowledge: **any button on the unit**, or **ACK** on the dashboard. Same code path.

Panel-side faults

LINK LOST	>15s with no frame — a **panel** fault. Check `solari-panel` on lithium and the USB cable
NO DATA	No snapshot yet, or an empty fleet. Normal for a few seconds after a restart
stale	Data older than 30s. The panel keeps rendering last-good on purpose; look upstream

Acknowledging silences; it never clears. The alert stays active in the pipeline — ack only records that a human saw the panel, and it is scoped to that alarm episode: a new episode re-arms tone, inlay and beacon from scratch. Note also that **while an alarm is armed every physical button is an acknowledge**, so pressing A for a theme change will silence the alarm instead.

The panel is the authority on its own state. A control on the dashboard is a request, not a switch: it shows pending until the panel reports back that it applied, then applied, or failed after 2 minutes with no confirmation. Nothing is shown as selected while a change is in flight, so a selected chip always means the panel said so. Expect 5–7s of latency, and don't re-click through it.

IF SOMETHING LOOKS OFF

Is the pipeline flowing?

On **xenon**:

```
systemctl is-active sor-emit sor-apply-dns sor-apply-pihole sor-reconcile notify
```

On **radium** (AD applier lives here):

```
systemctl is-active sor-apply-ad
```

Did a change reach DNS?

```
dig +short <name>.akoria.net @xenon
```

```
dig +short <name>.akoria.org @radium
```

Appliers are idempotent + self-reconcile (AD hourly), so drift self-corrects.

Is a host actually healthy?

Port-open isn't healthy (see the cesium incident). Check the per-host telemetry in **Assets → host**, or the underlying columns on `hostCurrent` (`fsReadonlyCount`, `blockDevMissing`, `smartFailCount`, `failedUnitCount`, `dmesgCritCount`). A real host-fault alert lands on your Apple devices via iMessage as crit/warn with the metric + detail in the body (e.g. `fsReadonly: /data (ro)`) — same channel as any other alert, plus a synthetic `node <X> stopped reporting` if the host goes silent instead of erroring.

Credentials never live in the repo. They're in `/root/.*-pw` on each host (and private agent memory). If a doc or config shows a password, that's a bug — tell me.

Deeper reading: `docs/SolariNet_RC_Handoff.md` (release-candidate handoff), `docs/SolariNet_Overnight_Build.md`, `docs/design/HOST_HEALTH_CONTRACT.md` (host-health + HTTP probe contract), `docs/SolariNet_Incident_2026-07-07_cesium.md` (the cesium post-mortem), `docs/panel/SolariNet_Panel_Manual.html` (the full status-panel manual — every screen, region by region), `docs/panel/SolariNet_Panel_Users_Guide.html` (the five-minute panel user's guide: buttons, screens at a glance, alarms, control page), and per-component READMEs under `deploy/notify/`, `deploy/sorsync/`, `netdb/sor/`. · Repo: github.com/soren42/Solarinet